

Correcting the transfer to the next winding without a human, and judging it by the text it carries

vesuve 0.3.0, a submission to the September 2026 Progress Prizes

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Abstract. Virtual unwrapping fits a surface to one sheet of a carbonised scroll inside an X-ray tomogram. When the surface slips onto the neighbouring winding, a human finds the switch and repairs it; the Vesuvius Challenge names these sheet switches a bottleneck and asks for conservative failure detection. We present **vesuve**, one program with one pipeline per prize, and three hand-free steps it performs on a published PHercParis4 segment. It **certifies** where the surface stayed on one winding, by closing loops on a lattice of steps between chunks: 6333 of 97771 chunks. It **produces** the next winding from the published surface prediction, by taking the first sheet along each normal and letting neighbours vote: 92.14 % of the points land on the right winding, against 76.13 % for a fixed step. It **corrects** the points that slipped, by modelling each chunk’s departure from its neighbourhood as noise or a one-winding slip: 163 misses become right for 41 rights made misses (sign test $p = 2.04 \times 10^{-18}$), and no judge takes part in any decision. Where the segment passes over the winding it produced, the published ink map says what text that winding must carry; our ink reading of the produced winding correlates with it at 0.83, against 0.12 and 0.10 for two controls, and at 0.87 one jump further. On a second band the correction does not hold, and the program measures that and writes nothing there. Every number is replayed by a test against the research function that produced it, and the inputs of the correction are rebuilt from public data byte for byte.

1 Introduction

A Herculaneum scroll is read in three stages: a tomogram of the closed scroll, a surface fitted to one sheet of papyrus inside it, and an ink model applied to that surface once flattened [1], [2]. The middle stage, segmentation, is where most human time goes. A surface that follows its sheet for a few centimetres and then slips onto the neighbouring winding produces a flattened image that mixes two texts, and today a human finds the switch and repairs it. The Challenge’s 2026 Open Problems page names sheet switches as a bottleneck and asks for “conservative failure detection”¹.

This report describes the parts of that job that **vesuve** does without a human, on real data, and where it stops. Its contributions are four:

1. a **certificate**: loops closed on a lattice of steps between neighbouring chunks say which chunks of a published surface stayed on one winding;
2. the **next winding**, computed from the published surface prediction rather than drawn by hand;
3. a **correction** of that next winding where a point slipped, decided without a judge;
4. a **judge by the text**: where the segment passes over the produced winding, its published ink map says what text that winding must carry.

Each was established by an experiment that declared its rule before measuring, and the negative results are reported alongside. The program ports only what was established, and its tests replay each ported number against the research code that produced it.

2 Related work

Recent Progress Prize work attacks the same failure while a surface is built. Stevens assembles unwrappings from patches and detects incompatible 3D locations, so that sheet switches are avoided during the assembly

¹https://scrollprize.org/2026_open_problems

[3]; ScrollFiesta extracts surfaces from the prediction cube by cube and welds them into sheets [4]. Both produce surfaces. **vesuve** starts from a surface that is already published and asks two other questions: which parts of it stayed on one winding, and how to repair the next winding with a decision no judge takes part in, whose rate of error is then measured. Neither its certificate nor its correction depends on how the mesh was made; we have not yet run them on meshes from these tools.

3 Data

All data are public and read from the Challenge’s bucket.

- **PHercParis4, segment 20230702185753**: its surface volume, read chunk by chunk; its published mesh; its published ink map; and, as the only judge, the next winding the segment’s own tracer drew by hand. For rendering, the raw scan the segment was cut from, at $2.4\ \mu\text{m}$ (20260411134726).
- **The surface prediction m7**, sampled along the normals of the mesh. The samples are embedded (0.5 MB, one bit per sample) and can be read again from the bucket (1780 chunks, about 630 MB).
- **PHercParis4, band 20260623142658-w028-037**: a second surface, where the correction was measured not to hold.
- **PHerc1447, segment 20250702235910**, for First Letters, and the innermost published PHercParis4 bands (w010-027), for the title search.

4 One program, one pipeline per prize

vesuve is a Python package over a small C core, shipped as one Docker image. Each prize has its pipeline, and each pipeline writes a dated report that says where it stops (Table 1).

pipeline	what it produces on real data, and where it stops
grand-prize	the per-chunk certificate (6333 of 97771 chunks), the certified surface, the next winding computed from m7, and that transfer corrected without a hand; it still has to read 111 bands (24263 chunks) to judge the rest
progress	where a published segment changes winding: column 260, rows 26 to 223, where the loop’s cumulative closure crosses half a sheet at cuts 163, 173 and 203
first-letters	a $4\ \text{cm}^2$ window chosen on papyrus alone, its fibre render and ink; on PHerc1447 the 2023 model shows no periodic rows at any angle, so it claims no letters
paris4-title	the last written column of the innermost band and the region after it, where an end-title would sit, for a papyrologist to judge

Table 1: The four pipelines of **vesuve** 0.3.0. Their dated reports are in `examples/`.

The rest of this report follows the **grand-prize** pipeline, which carries this month’s result.

5 Method

Terms. A scroll is one long sheet of papyrus rolled on itself; one turn of it is a **winding**. A **surface** is a mesh fitted to one winding inside the tomogram, made of points x_k with unit normals n_k . The **next winding** lies one turn further along the normal. A **transfer** gives each point a depth τ_k along its normal, so that $y_k = x_k + \tau_k n_k$ lies on the next winding; a point **slips** when y_k lands one winding too far or too short. Distances are in voxels of $v = 2.4\ \mu\text{m}$. The published sheet-to-sheet step is $s = 173\ \mu\text{m}$, and **half a sheet**, $\delta = \text{round}(s/(2v)) = 36$ voxels, is both the search range and the threshold of every decision below.

5.1 Closed loops certify where a surface stayed on one winding

The surface is flattened and cut into a grid of square **chunks**, in rows r and columns c . Where two chunks meet, along their **seam**, a **step** measures how far the sheet moves in depth from one chunk to the other: the depth profiles of the two chunks are aligned, and the offset that matches them is the step. Each seam is measured along k independent lines; its step is the median $c(s)$ of the line estimates $\hat{s}_\ell(s)$ that are present, kept only when a majority of the k lines are:

$$c(s) = \text{median}\{\hat{s}_\ell(s)\} \quad \text{if } \#\{\ell\} \geq \lfloor k/2 \rfloor + 1. \quad (1)$$

Write $H(r; c_0 \rightarrow c_1)$ for the sum of the steps along row r from column c_0 to c_1 , and $V(c; r_0 \rightarrow r_1)$ for the sum along column c . Around a rectangle of chunks, two paths lead from one corner to the opposite one, and their difference is the **closure** of the loop:

$$L = H(r_0; c_0 \rightarrow c_1) + V(c_1; r_0 \rightarrow r_1) - H(r_1; c_0 \rightarrow c_1) - V(c_0; r_0 \rightarrow r_1). \quad (2)$$

On a surface that stays on one winding the closure is zero up to the noise of the steps. A surface that switched winding inside the loop leaves a whole sheet in it, about 2δ . Along a row of loops, the cumulative closure $P_j = \sum_{i \leq j} L_i$ must stay within half a sheet, $|P_j| < \delta$ for every j . Chunks enclosed by loops that pass this test are **certified**. Where seams are missing, the program lists the bands of chunks it would still have to measure (Figure 1).

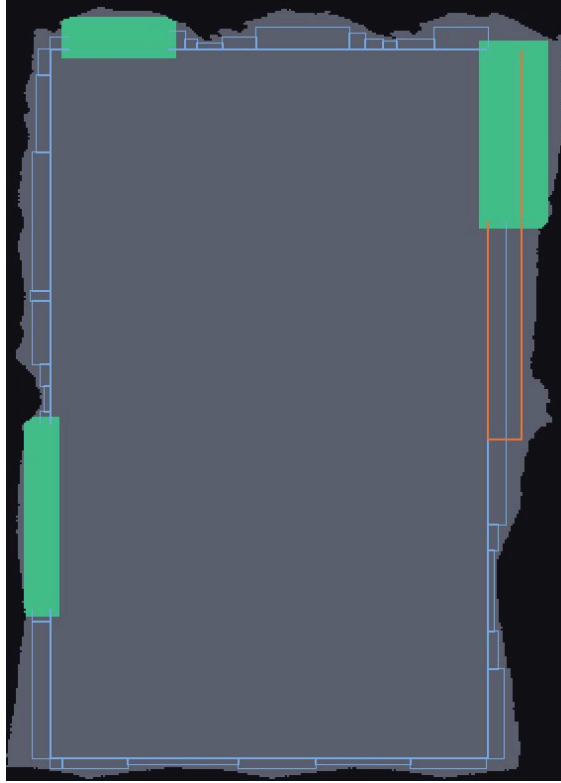


Figure 1: The certificate of segment 20230702185753, as `vesuve grand-prize` writes it: 6333 of 97771 chunks certified. Green: certified chunks; grey: present and not certified. Outlines: blue, the loops whose seams it would still have to measure; orange, a loop whose closure crosses half a sheet.

5.2 The next winding is the first sheet the prediction marks, settled by a vote

The published surface prediction P is a 3D map of where papyrus surfaces are, f times coarser than the scan. Along the normal of each point, from the surface out to three steps on one side, P marks runs of consecutive samples; the centre of each run is a sheet the next winding may land on:

$$C_k = \left\{ \frac{t_a + t_{b-1}}{2} : [a, b) \text{ a maximal run of } P \left(\left\lfloor \frac{x_k + t_i n_k}{f} \right\rfloor \right) > 0 \right\}, \quad t_i = \sigma i, \quad 0 \leq i \leq \lfloor 3s \rfloor. \quad (3)$$

The surface's own sheet is the run within 12 voxels of it. The first choice $\tau_k^{(0)}$ is the next run after that one. Then every point moves at once to the sheet its own ray sees nearest to its neighbours' median, within half a sheet:

$$\mu_k^{(r)} = \text{med} \left\{ \tau_j^{(r)} : j \in W_k \right\} \quad \text{if } |W_k| \geq 5, \quad \tau_k^{(r+1)} = \begin{cases} \text{argmin}_{c \in C_k} |c - \mu_k^{(r)}| & \text{if } \min_{c \in C_k} |c - \mu_k^{(r)}| < \delta \\ \mu_k^{(r)} & \text{otherwise} \end{cases} \quad (4)$$

where W_k is the three by three square of mesh cells centred on k . The vote stops when fewer than one point in a thousand moves by more than half a voxel, thirty rounds at most.

5.3 A slip is corrected only when it explains a point better than noise does

The correction needs no judge. First, a least-squares **walk** over the steps s_{ij} of the seams gives the depth D_c of the sheet in every chunk c , up to a common offset:

$$\hat{D} = \underset{D, \sum_c D_c=0}{\operatorname{argmin}} \sum_{(i,j)} (D_j - D_i - s_{ij})^2. \quad (5)$$

It is walked twice, on the surface the transfer starts from $(\hat{D}^r)$ and on the produced winding $(\hat{D}^p)$. Their difference is the transfer's depth chunk by chunk. The chunks are grouped into **blocks** of 16 by 16. Each block is given the level it should have from its neighbours only, so that its own slip cannot pull it: $a = \operatorname{median} \{ \hat{D}_c^p - \hat{D}_c^r : c \in \mathcal{N} \setminus B \}$, where B is the block and $\mathcal{N}$ its neighbourhood. The **departure** x of a point from that level is either noise or a slip of one winding above or below:

$$x \sim w_0 \mathcal{N}(0, \sigma^2) + w_+ \mathcal{N}(g, \sigma^2) + w_- \mathcal{N}(-g, \sigma^2), \quad (6)$$

with the weights and the shared width fitted by expectation-maximisation, and the size of a slip, $g = 69.458$ voxels, read from the data. A point is moved back by its departure, $\tau_1 = \tau_0 - x$, only when a slip explains that departure better than the noise:

$$\max(w_+ e^{-(x-g)^2/(2\sigma^2)}, w_- e^{-(x+g)^2/(2\sigma^2)}) > w_0 e^{-x^2/(2\sigma^2)}. \quad (7)$$

How it is scored. Only afterwards, the result is compared with the next winding as the segment's own tracer drew it by hand, alone and completed by three neighbouring segments; a point is scored only where these two judges agree within half a sheet. A moved point is **made right** when it was on the wrong winding before and on the right one after, and **made wrong** in the reverse case. With a points made right and b made wrong, the exact two-sided sign test asks how often chance alone, one in two, would split them at least as unevenly:

$$p = \min \left(1, 2 \sum_{i \leq \min(a,b)} \binom{a+b}{i} 2^{-(a+b)} \right). \quad (8)$$

5.4 The text the produced winding carries is a judge that knows nothing of the transfer

The segment makes more than one turn, so in places it passes over the very winding the transfer produced, one turn further along its own surface. There, its published ink map says what text the produced winding must carry. We read the ink of the produced winding, I , with the model that made the published map (`scrollprize/ink_canonical_2um`), and compare it with the published map J at the facing point $\varphi(u)$, on the mask where the segment passes within half a sheet, $M = \{u : |d(u)| < \delta\}$, d being the distance between the two surfaces:

$$r = \frac{\sum_M (I - \bar{I}) (J \circ \varphi - \overline{J \circ \varphi})}{\sqrt{\sum_M (I - \bar{I})^2 \sum_M (J \circ \varphi - \overline{J \circ \varphi})^2}}. \quad (9)$$

Three controls replace one term at a time: the text of the starting winding instead of the produced one, the facing point shifted by about one letter, and the winding one turn back. Six blocks were chosen from the meshes alone, before any ink was read on them.

6 Results

6.1 Computed from the prediction, the next winding is right for 92 % of the points

Computed from the prediction, the transfer lands on the right winding for 92.14 % of the points of the segment (Table 2). The program gives back the research's transfer point for point, within a millionth of a voxel, on the segment and on the second band.

rule	points on the right winding
every point moved by one fixed step	0.7613
first sheet along the normal, no vote	0.9119
first sheet along the normal, then the vote	0.9214

Table 2: Share of the transferred points of segment 20230702185753 that land on the right winding, judged by the winding its tracer drew by hand.

6.2 The correction is right four times as often as it is wrong, and says where it does not hold

On the 340 blocks of the segment that have neighbours along both axes, the correction moves 495 points: 163 are made right and 41 made wrong, a net gain of 122 (Table 3). The sign test gives $p = 2.04 \times 10^{-18}$ on the points, and $p = 2.25 \times 10^{-5}$ on the blocks, 42 up and 11 down. Replaced by noise, the judges change the counts and not one corrected point: they score, they do not decide.

setting	made right	made wrong	sign test p
segment, neighbours along both axes (340 blocks)	163	41	2.04×10^{-18}
segment, neighbours east and west only	net +5		
second band (84 blocks)	15	25	0.154
second band, neighbours along both axes	7	5	0.774

Table 3: The correction where it holds and where it does not. The second band is the published PHercParis4 surface 20260623142658-w028-037.

The gain has to be read at its real size. Most transferred points were already right, so over the whole segment the share on the right winding moves from 0.9303 to 0.9334. What matters is that the decision to move a point is taken without a judge and is right four times as often as it is wrong, which is the property a human corrector provides.

The negative rows are kept on purpose. With neighbours to the east and west only, even the segment’s gain falls from 122 to 5, and the level read from the neighbours carries the loss. Yet giving the second band’s blocks neighbours to the north and south does not bring its gain back. The program therefore writes corrections only for blocks with neighbours along both axes, and its report says that this condition is necessary and not sufficient.

6.3 The produced winding carries the segment’s text, one jump further too

The ink reading is first calibrated on the winding the tracer drew, where it correlates at 0.96 with the published map. On the produced winding, where the segment passes within half a sheet, it correlates at **0.83**, against 0.12 for the text of the starting winding and 0.10 for the shifted facing point; each of the six blocks alone lies between 0.68 and 0.92 (Table 4, Figure 2).

jump along the chain	r	controls
first	0.8331	0.1166 (starting text), 0.1037 (shifted facing point)
second	0.8749	0.1761, 0.0016, 0.2914 (one turn back)
third	0.4459	0.3992 at best: undecided

Table 4: The text judge along a chain of windings, each produced from the previous one. The second and third jumps are research, not yet in the program.

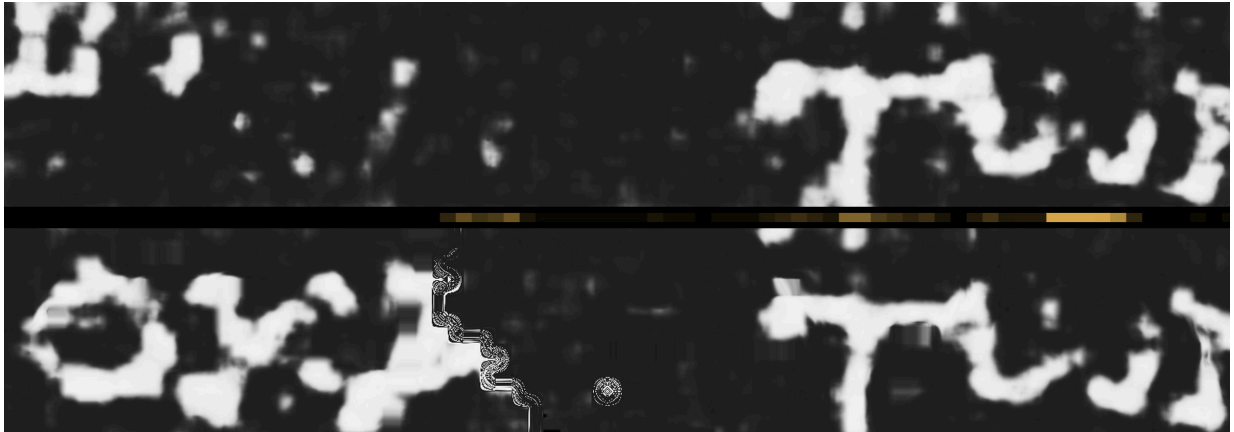


Figure 2: Top: the ink read on the produced winding, over a 29.5×4.9 mm strip of row 176. Bottom: the published ink map where the segment passes over that winding. In amber, where the segment passes within half a sheet, the only place where the comparison judges anything. This strip was looked at before the experiment was written; the six blocks behind the 0.83 were not.

The text judge sees little: a median of 2.2 % of a block lies within half a sheet of the segment. At the third jump it separates nothing, and we report it as undecided rather than as a pass.

7 Reproducibility

- **Published numbers, replayed.** Each port is tested against the research function that produced the number. The correction gives back what the research published on the segment and on the second band, block by block, and the corrected transfer it writes corrects the same points as the one the research saved, to within a millionth of a voxel.
- **Inputs, made again from public data.** `vesuve grand-prize --render` renders two piles per block through `vc_render_tifxyz` [5], from the published mesh and the raw scan. Over the whole segment (5 h 09 on three cores, 288 GB read), its 680 step tables equal the research's seam for seam, 334 171 seams with none different, and the corrected transfer it writes has the research's SHA-256.
- **Tests that can fail.** Rules were broken on purpose to check that their tests turn red: an anchor that keeps its own block, or a decision that favours the slip, makes four correction tests fail; a broken vote loses the right winding.
- **One command.** `docker run` reproduces the Grand Prize and Progress reports from the embedded data, offline: `uv run vesuve grand-prize --no-surface` takes about 20 s.
- **Every equation in one place.** `FORMULARY.md` is rendered from the code: each equation, the fact that carries it, and the research file that established it.

8 Limitations

`vesuve` does not unroll a scroll. It certifies, produces and corrects windings of surfaces others traced, on one segment of PHercParis4, which is not one of the thirteen scrolls of the 2027 Grand Prize. The correction is validated on that segment only: on a second band it does not hold, and the condition the program enforces (neighbours along both axes) is necessary, not sufficient. The text judge needs the segment to pass over the produced winding, which covers a small part of each block. The program does not read text: First Letters claims no letters on PHerc1447.

9 Conclusion

Three steps that a human performs when repairing a sheet switch (finding where a surface stayed on its winding, producing the next one, and bringing back the points that slipped) run here without a human, on public data, with decisions that are right four times as often as wrong and a judge that only scores. Where they do not hold, the program measures that and writes nothing. The next steps are the chain of windings, each jump judged by its text, and a scroll with no published trace.

References

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- [4] B. Kyles and others, “ScrollFiesta: A Virtual Meshing and Unwrapping Toolkit for the Herculaneum Papyri.” [Online]. Available: https://github.com/Hob3rMallow/scrollfiesta_public
- [5] Vesuvius Challenge, “villa: ScrollPrize monorepo (Volume Cartographer, VC3D, ink-detection, lasagna).” [Online]. Available: <https://github.com/ScrollPrize/villa>